

2. Off-grid applications covering behind-the-meter (BTM) systems are installed at the customer side of a utility meter and primarily help reduce costs and improve resiliency for commercial and industrial (C&I) or residential customers (Table II)

3. Remote power systems refer to storage systems operating as part of isolated electricity networks

Remote power systems are still a niche segment compared to grid-tied and off-grid applications.

Grid-tied scenario

Key applications at grid scale include provision of fast-response ancillary services such as frequency regulation, renewable ramp rate control, smoothening and firming up of renewable power as well as optimisation of transmission and distribution investments. Both policy makers and industry stakeholders are taking active steps for exploring energy storage integration in India.

Major policy boosts by Ministry of New and Renewable Energy (MNRE) and Power Grid Corporation of India Ltd (PGCIL) have opened doors to grid-tied energy storage deployment for ancillary services and renewable energy integration in India.

Telecom Regulatory Authority of India (TRAI) has mandated use of renewable power for telecom towers in India; currently, these towers use diesel power as the primary source of energy. These policy initiatives have resulted in tremendous opportunities for integration of energy storage technologies in a variety of applications. Immediate applications range from telecom tower backups to grid ancillary services and renewable integration.

Off-grid opportunities

Need for electrification and insufficient power supply are one of the primary growth drivers for the off-grid energy storage market in India. Electricity consumption is imperative for economic growth, driving infrastructure and social development. Grid extension has high operational cost and demands high investment. Consequently, in remote areas with limited grid connectivity, diesel gen-

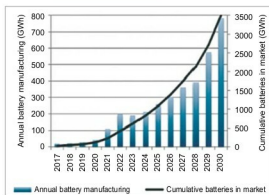


Fig. 2: Annual and cumulative battery requirements to meet India's EV ambitions (Source: NITI Aayog Report)

erators and energy storage systems play a crucial role. Technavio research estimates the off-grid energy market to grow steadily at a CAGR of more than 15 per cent by 2021.

The cost of diesel is higher in remote locations due to the added high cost of transportation. In the process of burning the diesel, diesel engines make noise and emit carbon into the atmosphere. Solar photovoltaic (PV) systems produce clean and renewable solar power while creating minimal noise during operation. The operating time of diesel generators can be reduced significantly by hybridising power systems, which will consequently reduce CO₂ emissions. According to Technavio study, demand for CO₂ emission reductions will gain traction in India's off-grid energy storage market.

The off-grid energy storage market in India is highly fragmented due to the presence of a large number of international, regional and local vendors. There will be fierce competition between big vendors offering advanced and differentiated products and smaller vendors offering high quality at lower prices.

Based on end use, the off-grid energy storage market in India can be segmented as residential and non-residential.

Majority of the population residing in India has limited access to grid power, which encourages consumers to adopt off-grid energy solutions such as solar home systems. For remote locations, solar home systems are economically viable and highly suitable to meet the growing demand for power,

contributing to the growth of residential solar energy storage market.

Electric vehicles to drive energy storage

India has launched a National Mission on Electric Mobility with a target of six million electric vehicles (four million two-wheelers and two million four-wheelers) by 2020. Efficient rollout of the EV programme requires electrical distribution infra-

structure upgrades and smarter systems to control/limit simultaneous charging of hundreds of EVs from the same feeder.

Further, India's aspiration to achieve 100 per cent electric vehicle (EV) sales by 2030 will drive up its rank among the top countries manufacturing batteries. However, this will require a strategy to overcome India's relatively weak initial position in battery manufacturing while claiming an increasing share of total battery value over time. India's EV battery market alone could be worth as much as \$300 billion from 2017 to 2030. The country could represent more than one-third of global EV battery demand by 2030, provided it meets the goal of a rapid transition to shared, connected and electric mobility (Fig. 2).

Since battery accounts for about one-third of the total purchase price of an EV, driving down battery costs by rapidly scaling up production and standardising battery components could be the key to long-term success for India's automotive sector. In fact, India's EV mission could drive down global battery prices by as much as 16 per cent to \$60 per kWh.

Immediate policy-level support is needed to build the enabling infrastructure for integration of EVs in the electrical network so that these millions of EVs connected to the power system can be leveraged as virtual power plants that store surplus energy generated and support the grid during moments of deficit. Globally, Vehicle to Grid (V2G) technologies are evolving rapidly to achieve these objectives.